Hands-On-Learning: Design & Secure LLM APIs for Scalability

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Description

GlobalCorp, a Fortune 500 enterprise, is launching an LLM-powered customer support API that will serve ten million users across fifty countries. The platform must handle one hundred thousand requests per minute, maintain 99.9 percent uptime, meet HIPAA and SOC 2 compliance, and stay within a five-hundred-thousand–dollar monthly budget.

Your role is to design a production-ready LLM API platform that integrates microservices architecture, enterprise-grade security, and operational excellence.

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Learning Objectives

By completing this project, you will demonstrate:

- Ability to design scalable microservices architectures with API Gateway patterns.
- Proficiency in implementing multi-layered security with OAuth2, JWT, and prompt-injection defenses.
- Competence in building observability stacks and cost-optimization strategies.

Project Components

Tools to be used

- API Gateway
- OAuth2 and JWT
- Logging and audit systems
- Monitoring tools (Prometheus, Grafana)
- Caching tools (Redis)
- Auto-scaling infrastructure
- Database tuning tools
- Observability stack

Scenario

GlobalCorp is launching an LLM-powered customer-support API with strict performance, compliance, and reliability requirements.

- The platform must:
- Support a massive, global user base.
- Maintain 99.9 percent uptime.
- Meet HIPAA and SOC 2 compliance.
- Stay within a fixed infrastructure budget.

You are the lead platform architect responsible for designing a production-ready LLM API platform with scalable architecture, multi-layered security, monitoring, and cost-optimization strategies.

Instructions for Learners

Step 1 — Analyze the Scenario

- Read GlobalCorp's API requirements (scalability, compliance, uptime, cost).
- Identify the core architectural, security, and operational challenges.

Outcome: You understand the system requirements and the constraints of the project.

Step 2 — Complete Deliverable 1: Microservices Architecture Design

1. Identify four to five microservices based on business domains.
2. Define each service's boundaries and responsibilities.
3. Map API Gateway routing rules and rate-limiting strategies.
4. Assign a database-per-service model and consistency strategy.
5. Create a high-level architecture diagram.

Outcome: A complete microservices architecture document with diagram.

Step 3 — Complete Deliverable 2: Security Architecture & Compliance

1. Define authentication flows using OAuth2, JWT, and API keys.
2. Design prompt-injection defense layers.
3. Configure audit logging and security monitoring components.
4. Map HIPAA and SOC 2 requirements to your architecture.

Outcome: A comprehensive security and compliance architecture.

Step 4 — Complete Deliverable 3: Monitoring & Performance Optimization

1. Define Prometheus metrics for request rate, latency, error rate, and cost.
2. Build Grafana dashboard layout with SLO indicators.
3. Propose caching using Redis with TTL and warming policies.
4. Document performance-enhancement strategies (indexing, read replicas).

Outcome: A monitoring and optimization plan ready for production.

Step 5 — Complete Deliverable 4: Cost Optimization & Disaster Recovery

1. Identify cost-reduction strategies (auto-scaling, model tiering, caching).
2. Estimate expected cost savings percentage.
3. Define RTO (≤ five minutes) and RPO (zero) for disaster recovery.
4. Create DR restoration runbooks and cost-monitoring workflows.

Outcome: A complete cost-optimization and DR strategy.

Compile Your Findings

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Requirement 1: Microservices Architecture Design

Description: Provide a complete microservices architecture for GlobalCorp's customer support API.

Include:

- Service boundaries
- API Gateway routing
- Database-per-service architecture
- Architecture diagram
- AI grading prompt

Requirement 2: Security Architecture and Compliance

Description: Provide a full security design with compliance mapping.

Include:

- OAuth2, JWT, API keys
- Prompt-injection defense
- Monitoring and logging
- HIPAA + SOC 2 checklist
- AI grading prompt

Requirement 3: Monitoring & Optimization

Description: Provide a monitoring, observability, and performance-optimization strategy.

Include:

- Prometheus metrics
- Grafana dashboards
- Caching policies
- Optimization techniques
- AI grading prompt

Requirement 4: Cost Optimization & Disaster Recovery

Description: Provide a cost-reduction and DR plan meeting GlobalCorp's reliability needs.

Include:

- Auto-scaling
- Model tiering
- Expected cost savings
- RTO/RPO targets

- DR runbooks
- Budget monitoring
- AI grading prompt

Summary

This project integrates microservices architecture, multi-layered security, enterprise operations, cost optimization, and real-world compliance into a unified LLM platform design for GlobalCorp.

You now have a complete production-ready strategy demonstrating architecture, security, monitoring, and cost-efficiency skills.

Resources

None required

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the "Peer Review" area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.